Boltzmann Wealth Model

Analytical solution in the limit $N \to \infty, \; T \to \infty$

Conservation | Max Entropy | Boltzmann–Gibbs

The Model & its Conserved Quantity

What stays fixed determines everything

Setup

1. N agents, each with integer wealth $w_i \geq 0$
2. Initial condition: $w_i = 1$ for all i
3. At each step: pick i and j at random;
   if $w_i > 0 \rightarrow w_i - 1,\ w_j + 1$
4. Repeat for T steps

Conserved quantity

$$W = \Sigma_i w_i = N = \text{const}$$

Total wealth is conserved at every step —
each transaction transfers, never creates or destroys.

$$\langle w \rangle = W / N = 1$$

Mean wealth is a constant of motion.
It plays the role of temperature.

Analogy: microcanonical ensemble with fixed total energy $E = N$

Derivation via Maximum Entropy

The stationary P(w) is the least-biased distribution consistent with constraints

Constraints on P(w)

$$(1)\quad \Sigma_w P(w) = 1 \qquad \text{normalization}$$

$$(2)\quad \Sigma_w w P(w) = \langle w \rangle \qquad \text{fixed mean}$$

Objective

$$\max\ S[P] = -\Sigma_w P(w) \ln P(w)$$

Lagrange multipliers

Setting $\delta L/\delta P(w) = 0$ with Lagrangian:

$$L = -\Sigma P \ln P - \lambda(\Sigma P - 1) - \beta(\Sigma w P - \langle w \rangle)$$

gives:

$$P(w) = Z^{-1} e^{(-\beta w)}$$

Normalization: $Z = 1/\beta = \langle w \rangle$ (continuous limit)

Constraint (2): $\beta = 1/\langle w \rangle$

This is the unique maximum-entropy distribution on [0, ∞) with fixed mean.

Connection to Statistical Mechanics

The wealth distribution IS a Boltzmann-Gibbs distribution

Formal mapping

Boltzmann-Gibbs	Boltzmann Wealth Model
Particle energy ε	Agent wealth w
Total energy $E = \text{const}$	Total wealth $W = N = \text{const}$
Temperature $k_B T$	Mean wealth $\langle w \rangle$
$P(\varepsilon) = (1/Z)\, e^{(-\varepsilon / k_B T)}$	$P(w) = e^{(-w/\langle w \rangle)} = e^{(-w)}$

Key insight: Random pairwise exchange conserving total wealth is the wealth-transfer analogue of elastic collisions in an ideal gas. Both systems thermalize to the same maximum-entropy distribution.

The Stationary Distribution

Continuous approximation and exact discrete result

Continuous limit ($N \to \infty$)

With $\beta = 1/\langle w \rangle$ and $\langle w \rangle = 1$:

$$P(w) = e^{(-w)} \qquad w \geq 0$$

Mean: $\langle w \rangle = 1$ ✓

Variance:

$\mathrm{Var}(w) = 1$

Lorenz curve:

$$L(p) = p + (1-p)\ln(1-p)$$

The Lorenz curve answers this concrete question:
If I order all agents from the poorest to the richest, what fraction of the total wealth is owned by the poorest p%?

Discrete (exact, integer wealth)

For integer $w \in \{0, 1, 2, \ldots\}$ with $\langle w \rangle = 1$, the exact solution is geometric:

$$P(w = k) = (1/2)^{(k+1)}$$

Explicit values:

k	0	1	2	3	4
P(k)	1/2	1/4	1/8	1/16	1/32

Both converge for large k: $(1/2)^{(k+1)} \approx e^{-(k \ln 2)} \approx e^{-(0.693\,k)}$

Inequality: the Gini Coefficient

A quantitative prediction from $P(w) = e^{-w}$

Definition

$$G = 1 - 2 \int_0^1 L(p)\, dp$$
$$L(p) = p + (1-p)\ln(1-p) \quad \text{(Lorenz curve)}$$

Result for the exponential distribution

$$\begin{aligned} G &= 1 - 2 \int_0^1 [p + (1-p)\ln(1-p)]\, dp \\ &= 1 - 2\left[\tfrac{1}{2} + \int_0^1 (1-p)\ln(1-p)\, dp\right] \\ &= 1 - 1 - 2\left(-\tfrac{1}{4}\right) = \tfrac{1}{2} \end{aligned}$$

$$\boxed{G = 1/2}$$

Exact, independent of $\langle w \rangle$. Valid for any exponential distribution.

Interpretation

G = 0

perfect equality
(everyone has the same wealth)

G = 0.5

the BWM prediction:
the bottom 50% own ≈ 15% of total wealth

G = 1

maximum inequality
(one agent owns everything)

Why does inequality emerge from perfect equality?

An entropic argument — no mechanism needed

The question

All agents start with $w_i = 1$.
The rule is symmetric: no agent has any advantage.
Yet $G \to 1/2$. Why?

Counting microstates

The uniform state $\{1, 1, 1, \ldots, 1\}$ is a single microstate. There are vastly more microstates that look like an exponential distribution than those that look like the uniform one.

The analogy

Identical to a gas of N molecules with fixed total energy E = N. Equipartition is one state; Boltzmann is the overwhelmingly typical state.

Formal statement

$P(w) = e^{-w}$ is the unique distribution that maximises Shannon entropy subject to $\langle w \rangle = 1$.

$$S[P] = -\int P(w) \ln P(w)\, dw \;\to\; \max$$

Punchline

G = 1/2 is not a sign of injustice — it is the thermodynamic equilibrium of random pairwise exchange. The initial uniform state is extraordinarily improbable and is immediately destroyed by random dynamics.

G = 1/2 is the signature of maximum entropy, not of an unfair rule.

Deriving the Lorenz Curve

What fraction of total wealth is held by the bottom p% of agents?

Definition of L(p)

L(p) = fraction of total wealth held by the poorest fraction p.

$$L(p) = \frac{\int_0^{w^*(p)} w\,e^{-w}\,dw}{\int_0^{\infty} w\,e^{-w}\,dw} \quad (\text{denom.} = \Gamma(2) = 1)$$

Step 1 — find w*(p) from the CDF

Cumulative distribution: $F(w) = 1 - e^{-w}$

$$F(w^*(p)) = p \implies w^*(p) = -\ln(1-p)$$

Step 2 — evaluate the denominator

$$\Gamma(2) = 1$$

Integration by parts: numerator

Compute $\int_0^{w^*} w\,e^{-w}\,dw$. $u = w,\ dv = e^{-w}dw$
$du = dw,\ v = -e^{-w}$

Step 1 — apply ∫ u dv = [uv] − ∫ v du

$$= \left[w(-e^{-w})\right]_0^{w^*} - \int_0^{w^*} (-e^{-w})\,dw$$

Step 2 — evaluate the boundary term [uv]$_0$^{w*}

$$= \left[-w\,e^{-w}\right]_0^{w^*} + \int_0^{w^*} e^{-w}\,dw$$

$$= -w^*e^{-w^*} + (0-0) + \left[-e^{-w}\right]_0^{w^*} \quad (\text{at } w=0\text{: } 0\cdot e^0 = 0)$$

Step 3 — evaluate [−e^{−w}]$_0$^{w*} = −e^{−w*} + 1

$$= -w^*e^{-w^*} - e^{-w^*} + 1 = 1 - (1 + w^*)e^{-w^*}$$

$$\int_0^{w^*} w\,e^{-w}\,dw = 1 - (1 + w^*)e^{-w^*}$$

$$\boxed{L(p) = p + (1-p)\ln(1-p)}$$

The Gini Coefficient $G = 1/2$

Area between the diagonal and the Lorenz curve, normalized

Geometric definition

G measures how far L(p) is from the equality line $L(p) = p$:

$$G = 2\int_0^1 [p - L(p)]\,dp = 1 - 2\int_0^1 L(p)\,dp$$

Since $\int_0^1 p\,dp = 1/2$ gives the triangle area = 1/2.

Properties of L(p)

- $L(0) = 0$, $L(1) = 1$ (boundary conditions)
- $\ln(1-p) \le 0$ for $p > 0 \implies L(p) \le p$ always
- concave: poor agents own a smaller share

Calculation

$$G = 1 - 2\int_0^1 [p + (1-p)\ln(1-p)]\,dp$$

$$= 1 - 2I_1 - 2I_2$$

where:

$$I_1 = \int_0^1 p\,dp = 1/2$$

$$I_2 = \int_0^1 (1-p)\ln(1-p)\,dp$$

Substitute $u = 1-p$ in I_2:

$$I_2 = \int_0^1 u \ln u\,du = [u^2 \ln u / 2]_0^1 - \int_0^1 u/2\,du$$

$$= 0 - 1/4 = -1/4$$

(the boundary term $u^2 \ln u \to 0$ as $u \to 0^+$)

$$G = 1 - 2(1/2) - 2(-1/4) = 1/2$$

$L(0.5) = 0.5 + 0.5\ln(0.5) \approx 0.153$ → the bottom 50% own only ~15% of total wealth. $L(0.8) \approx 0.558$ → the bottom 80% own ~56%, so the top 20% own ~44%.

Finite-N and Finite-T Effects

How fast does the simulation converge to the analytical limit?

Relaxation time τ

Each agent is selected once every ~N steps on average.
The system needs $T \gg N$ steps to reach stationarity.
Practical rule: $T_{min} \approx 100\,N$

Convergence criterion

Compare empirical $\hat{P}(w)$ to e^{-w} using KL divergence or Kolmogorov–Smirnov test at intervals of N steps.

Initial transient

The first ~N steps are dominated by initial conditions.
Discard the transient before computing averages.

Finite-N fluctuations

Statistical fluctuations in $\hat{P}(w)$ scale as $1/\sqrt{N}$.
For N = 100: ~10% noise. For $N = 10^4$: ~1% noise.

Wealth discretisation

For small N the geometric (discrete) distribution is more accurate than the continuous exponential.

Practical guide

N = 50 · T = 5 000 → visible exponential, noisy

N = 500 · T = 50 000 → clean fit to e^{-w}

N = 5 000 · T = 500 k → excellent agreement, G ≈ 0.50